

CASPER@night — Agilent/VnmrJ Quickstart

(SIU Carbondale — 400 MHz DD2, VnmrJ 3.2)

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1 What this is

This guide covers one specific session: a receive-only spin-noise measurement on the SIU Carbondale 400 MHz Agilent DD2 (VnmrJ 3.2, 5 mm HCN probe, room temperature). It is the Agilent leg of the CASPER@night network — the same physics protocol the Bruker fleet runs, recording the spin-noise feature of water on a room-temperature probe (the Guéron absorption dip in equilibrium; the September sessions here showed a +27% emission bump instead, which is why the tuning ladder of 2.6 now exists) — acquired by hand with ordinary VnmrJ tools (`s2pul`, `go`, `svf`) and converted into a network bundle afterward on any laptop. Nothing in the session pulses the sample above tune-and-match power: the calibration steps use $\sim 1^\circ$ tips at your ordinary observe power, and the noise blocks use `pw=0` at the bottom of the transmitter power scale. One thing to know going in: **this path has run on real hardware on one instrument — this one.** Two sessions on 2026-09-14 (68 and 66 experiments) were packed, validated and uploaded with zero reader warnings, so the converter has parsed real files from one instrument of this console family and no other. If your files make it choke, that is still data we need — ugly output is useful output, and Section 4 says what to send. You never need the GitHub repository, git, or a README: this PDF plus the `config.json` attached to the same e-mail is everything.

What you need

1. **Any laptop or desktop with Python 3.6 or newer** — macOS, Linux, and Windows all work; nothing is installed on the spectrometer. Check with:

```
python3 --version
```

2. **The software — one download, no git.** Open

```
https://github.com/CASPER-Night/spin-noise-network/releases/latest
```

and under *Assets* click **Source code (zip)**. Unzip it anywhere.

3. **The config.json from John's e-mail** — drop it into the unzipped folder at exactly `uploader/config.json` (next to `upload_bundle.py`). It is your facility's upload key — treat it like a password.
4. **A 5 mm tube of 90% H₂O / 10% D₂O, lightly doped** — a retired shim/lineshape sample is fine, as long as you know (or can best-guess) the composition. The H₂O fraction *is* the measurement.
5. **About 1 h 20 min of instrument time** — roughly 20 min of setup and calibration you attend, ≥ 30 min of noise blocks the console mostly runs itself, then some 15 min for the tuning ladder and the axis-sign check (2.6–2.7; about 25 min with the six-block ladder a 1%-class tube needs).

2 At the spectrometer

Everything below is standard VnmrJ operation — no macro, nothing installed on the console. Save every experiment with `svf` into one session directory, one `.fid` per step; the leading number in each save name identifies the experiment and the `_sn...` part tells the converter its role, so keep both as printed. Anything after them is free as long as it does not run straight on in letters: `17_sn_noise_b` and `17_sn_noise2` are noise blocks, `17_sn_noisetest` is not recognised and the converter stops and says so.

2.1 Setup (nothing new here, ~15 min)

Insert the water tube, set or record the temperature (298 K if VT is off and the bore is at room temperature), and let it equilibrate a few minutes. Tune and match ¹H by your normal manual wobble procedure — no autotune on this probe, and none needed. Shim as usual, with one caution. On a near-neat water tube the water line may sit at ~30 Hz wide and asymmetric, and gradient shimming cannot narrow it — that is radiation damping, the probe's back-action on the spins, not a shim fault. On this instrument, identical shims and identical acquisition gave an 8.5 Hz symmetric line on a 1% H₂O/D₂O Gd-doped tube and a 32 Hz asymmetric line on the 90/10 tube; three gradient-shim iterations on the 90/10 tube moved z3–z5 by several hundred DAC units each and changed the linewidth by under 0.2 Hz. Judge shim quality on the lock signal or on a dilute/doped tube, not on the near-neat water line, and do not judge a shim set by the size of its DAC values — the set that gave the 8.5 Hz line had z3 and z3y past ± 8000 . The broad line is the regime that produces a large noise feature, so leave it as it is.

Lock: run with the lock **on** during the noise blocks unless you have a reason not to. On this instrument the locked session drifted 0.25 Hz between opening and closing reference over a 38 min bracket, with an unchanged lineshape; the unlocked session drifted 0.56 Hz over a 78 min bracket.

That is 0.0065 versus 0.0072 Hz/min — the same drift rate within what two brackets can show — so the two sessions do not by themselves measure what the lock buys. (`alock='n'` in the block parameters below only disables automatic re-locking; it does not release an established lock.) Either way, **record the state** — the questionnaire asks for it. Calibrate the ^1H 90° pulse or take the probe-file value, and **write down pw90 and tpwr** — the questionnaire wants both, and the closing reference must reuse them. Start from your standard ^1H water setup in `s2pul` and leave `sw` at your usual value throughout.

Probe temperatures. The questionnaire (3.2) asks for the coil and preamplifier temperatures, and it means *physical* temperatures: on a room-temperature probe the coil sits at the probe-body temperature, which is the lab ambient, and the preamplifier at its own physical ambient. Read the lab thermometer (or the bore/VT readout with VT off) while you are at the magnet and enter that — a measured 296.5 K beats an assumed 298 K; `null` is for having no reading at all. If a better number turns up after the upload, e-mail it to the maintainer: corrections are applied at analysis time, as numbers, never by re-uploading.

2.2 Gain ladder (~3 min)

A randomized, repeated ladder of one-scan tiny-flip 1Ds — the receiver-linearity calibration. First find your ladder maximum: the highest **gain** at which a tiny-flip 1D of this tube does not overflow the receiver. **It is strongly sample-dependent** — on this probe 60 dB was fine on a 1% H_2O Gd-doped tube and the 90/10 $\text{H}_2\text{O}/\text{D}_2\text{O}$ tube needed 40 dB. The ladder then visits every level from 0 to that maximum in 5 dB steps (legal gains are integer dB: 0, 5, ..., 40 for a 40 dB maximum, nine levels; thirteen for 60 dB), **each level three times, in one random order**, with `pad=1` so the receiver settles for a second after every gain change. The September sessions used four ascending gains, and in both the rungs' amplitude per unit gain fell by about 10% from the bottom rung to the top one (11% and 10%) — which a monotonic ladder cannot attribute: receiver compression and signal drift are degenerate when gain and time rise together. Visiting each level three times at random moments separates them: compression follows the gain, drift follows the clock. Twenty-seven acquisitions of a few seconds each, about 3 minutes in all.

Set `pw` to `pw90/90` — 0.11 μs for a 10 μs `pw90`, about a 1° tip — and `tpwr` to your calibrated observe power (57 below stands in for yours). Write both values down: the references in 2.3, 2.5 and 2.7 must use exactly these. Never use `gain='n'` (autogain), and wait for each acquisition to finish before saving. The console may store a different gain than you typed — asked for 13.3 and 26.7 dB, this console stored 14 and 26 — which is why the converter reads the stored gain from each rung's `procpa` and you list nothing. Save the rungs under four-digit numbers counting up from 1000 in the order acquired (a number that can never collide with a noise block); the gain in the name is for your eyes only. One random order for a 40 dB maximum, drawn once and printed here so you need not draw it at the console — any other random order is as good, and the order that ran is on record in each file's time stamp:

save as	1000	1001	1002	1003	1004	1005	1006	1007	1008
gain	35	35	5	40	15	30	10	15	40
save as	1009	1010	1011	1012	1013	1014	1015	1016	1017
gain	10	25	30	20	5	40	30	35	0
save as	1018	1019	1020	1021	1022	1023	1024	1025	1026
gain	20	0	0	10	15	25	5	25	20

The first three rungs, then, are:

```

pw=0.11
tpwr=57
nt=1
pad=1
gain=35
go
svf('1000_sn_ladder_35')
gain=35
go
svf('1001_sn_ladder_35')
gain=5
go
svf('1002_sn_ladder_05')

```

and so on down the table. For a 60 dB maximum use the levels 0, 5, . . . , 60 (13 levels, 39 rungs, about 4 minutes), each three times, in an order you shuffle yourself before you start — write it down first. Read every **gain=40** below as your ladder-maximum value.

2.3 Opening reference (~1 min)

One more tiny-flip 1D at the ladder-max gain, with a 2 s record. **pw** and **tpwr** are the two values you wrote down in 2.2 (the same stand-ins below), and **pad** goes back to 0 — the ladder left **pad=1** behind:

```

pw=0.11
tpwr=57
pad=0
at=2
gain=40
go
svf('11_sn_ref_open')

```

2.4 Noise blocks (≥ 30 min — the point of the session)

No pulse at all — these lines are the receiver-only idiom of Agilent’s own **cryo_noisetest** macro, with **tpwr=-16** the bottom of the standard transmitter power scale:

```

pw=0
tpwr=-16
nt=1
ss=0
pad=0
wshim='n'
alock='n'
gain=40
at=30
go
svf('12_sn_noise')

```

Then repeat until you have at least 30 minutes of noise data in total — about 60 blocks at **at=30** — counting up from 17 (13 is reserved for the closing reference):

```

go
svf('17_sn_noise')

```

```
go
svf('18_sn_noise')
```

Three notes — two settled on this console, one still open:

- **Record length.** The console caps a record at a point count, $np \leq 524288$, so the longest record is $at_{\max} = 524288/(2\,sw)$ — about 41 s at $sw = 6410$ Hz. $at=30$ with about 60 blocks gives 30 minutes; both sessions here used 30 s records (62 and 60 blocks). A narrower sw buys a proportionally longer record — 3205 Hz allows about 82 s — if your usual water window permits it.
- **Transmitter silence.** Whether the DD2's transmitter chain is measurably silent at $pw=0$ $tpwr=-16$ has not been checked — no spectrum analyser was available on 2026-09-14. If the noise floor shows something odd — a carrier spike, a ridge — do not chase it: finish the session and describe it in the questionnaire notes.
- **Save names.** `svf` accepts names starting with a digit on this console — both sessions saved every experiment under the `NN_...` pattern. The leading number and the `_sn_noise` part are parsed; anything after them is free unless it runs straight on in letters (`17_sn_noise_b`, `17_sn_noise2` and `17_sn_noise-1` are read as noise blocks; `17_sn_noisetest` is not).

2.5 Closing reference (~1 min)

Repeat 2.3 exactly — same `at`, same `gain`, and `pw` and `tpwr` back at the two values you wrote down in 2.2 (the noise blocks left $pw=0$ $tpwr=-16$ behind, so both must be restored; do not retype the stand-in numbers below unless they are yours). The two references are a matched drift bracket, and a closing reference at a different flip angle cannot be compared with the opening one — the first session here took them at $pw=0.05$ and $0.1125\,\mu\text{s}$, a $2.25\times$ mismatch in flip angle, for exactly this reason. Save as 13:

```
pw=0.11
tpwr=57
at=2
gain=40
go
svf('13_sn_ref_close')
```

That is the main session; 2.6 and 2.7 add two short calibration series to it. An external acquisition driver — a Python 2.6 script that runs on the console workstation itself, submits one block at a time to a running VnmrJ through `listenon/send2Vnmr`, and waits for each `.fid` to land before submitting the next — ran the second session here end to end on 2026-09-14 (66 of 66 blocks, unattended). It is being contributed to `vendors/agilent/`; until it ships there, the manual steps above are the recommended path. The draft VnmrJ macro (`spin_noise_run.mac`) has still not run on hardware.

2.6 Spin-noise tuning ladder (15–25 min)

The sign and shape of the spin-noise feature depend on how the probe is tuned. Equilibrium physics expects the Guéron absorption dip on a room-temperature probe; both September sessions on this probe showed a clear feature of the opposite sign — an emission bump, +27% of the floor on the 1% H_2O tube — which says the probe was tuned away from the spin-noise tuning optimum, a setting that is in general not the wobble minimum you tune to. Stepping the tuning changes the shape of the feature as well as its size: at the optimum the dispersive admixture vanishes and the feature is the purest absorptive dip; away from it a dispersive component grows, and at the setting where the feature nulls and flips from bump to dip the absorptive part is passing through zero, so what remains

is nearly pure dispersion — that crossover is a setting *far* from the optimum, not the optimum itself. The report fits every setting with the same absorptive-plus-dispersive model as the main block, ranks the settings that reach 5σ by their dispersive fraction $|b/a|$, and names the smallest as nearest the optimum; the null-or-flip setting is the one it ranks farthest, so do not label it the optimum in your notes. So after the closing reference, step the tuning deliberately and record noise blocks at each setting: **at least five settings bracketing your normal one** — the normal tuning, then one and two steps to either side, where a step is whatever you can set and read reproducibly (a number of turns of the tune or match capacitor, a reflected-power reading on the wobble display, a wobble-curve minimum in dB). Same no-pulse parameters as 2.4, **at=30**, two blocks per setting on the recommended 90/10 tube; setting 0 is your normal tuning, saved as 2000 and 2001:

```
pw=0
tpwr=-16
nt=1
ss=0
pad=0
wshim='n'
alock='n'
gain=40
at=30
go
svf('2000_sn_tune_0')
go
svf('2001_sn_tune_0')
```

Then retune to setting 1 and save 2010_sn_tune_1 and 2011_sn_tune_1, setting 2 as 2020/2021, and so on: setting k goes to 20 k 0 and 20 k 1, and the trailing $_k$ tells the converter which setting a block belongs to. **How many blocks per setting depends on the tube.** The report claims a feature only at 5σ and ranks only settings that reach it. On this instrument the 90/10 tube gave about 11σ per 30 s block, so two blocks are plenty; the 1% H₂O Gd-doped tube gave about 2.5σ per block — two blocks reach 3.5σ , every setting comes out “undetermined”, and four reach 5σ only just. On a 1%-class tube take six blocks per setting (20 k 0 to 20 k 5, about 25 min for five settings). Do the normal setting *within* the ladder, as its own blocks, so the ladder carries its own anchor even though 2.4 was acquired at the same tuning. For every setting, **write down whatever the console shows** for tune and match — the readings go into the **tuning** entries of the questionnaire (3.2), in whatever units you have. These blocks are analysed as a ladder of their own and are **never averaged into the main noise block** of 2.4. Retune to normal when you are done.

2.7 Axis-sign check (~1 min)

On this console a positive FFT-convention offset points the opposite way from the physical one: the axis is *mirrored*, the physical line frequency is carrier $-f_{\text{FFT}}$. That was established on the partner console two independent ways (vendor checklist item 2): a +200 Hz **tof** step moved the apparent line by +199.6 Hz, and VnmrJ’s lock referencing (**reffrq**, the 0 ppm frequency) places water ~ 500 Hz *below* the carrier where the FFT convention had it ~ 540 Hz above. **tof** itself is recorded verbatim as **o1_hz**, no sign flip. The report still resolves the sign for every bundle instead of assuming it: from the one extra acquisition below when it is present, otherwise from the **procpa**r’s own referencing (**reffrq**, **rfl**, **rfp** against the water shift), and it flags a disagreement between the two. The acquisition is the reference of 2.5 once more, identical in every parameter, with the transmitter offset moved by exactly +200 Hz. With the probe back at its normal tuning:

```
pw=0.11
```

```

tpwr=57
pad=0
at=2
gain=40
tof=tof+200
go
svf('3000_sn_signal')
tof=tof-200

```

The saved `procp` carries the displaced `tof`; the converter records it, and the report takes `tof(sign check) - tof(reference)` as the displacement. In the physical convention the water line's apparent offset in the window would *drop* by exactly that much — 200 Hz; on this console it *rises* by 200 Hz, which is the mirrored axis, and the report states which it found. The last line puts `tof` back so nothing you run next inherits the shift.

3 Back at your desk

3.1 Copy the data

Copy the whole session directory — every `NN_*.fid` folder — onto the laptop, any path. A USB stick is fine; the console never needs internet access. Copy the console's `/vnmr/vnmrrev` (three lines of text) into the top of that same directory as `vnmrrev`. The packer takes the VnmrJ version from the `"vnmrj_version"` line of `answers.json` when that line is present — the template below pre-fills SIU's — and reads this copy only when the line is left out, so the copy is the fallback that lets you drop the line rather than retype the version.

3.2 Fill in `answers.json`

In the unzipped `spin-noise-network` folder, create `answers.json` with the content below — it is pre-filled for SIU, so only the FILL IN lines (and `"locked"`, `"p90_us"`, and `"vt_setpoint_k"` if yours differed) are yours to edit:

```

{
  "vendor": "agilent",
  "run_mode": "external-acquisition",
  "facility": {
    "institution": "Southern Illinois University",
    "city": "Carbondale, Illinois",
    "country": "USA",
    "facility_slug": "siu-carbondale",
    "contact_email": "bgoodson@chem.siu.edu",
    "contact_consent": true
  },
  "sample": {
    "description": "FILL IN (e.g. retired lineshape sample, 90% H2O / 10% D2O, Gd-doped)",
    "h2o_fraction_pct": 90,
    "d2o_pct": 10,
    "additives": "FILL IN (dopant, if known)",
    "tube_od_mm": 5,
    "sample_volume_ul": 550,
    "vt_setpoint_k": 298
  },
  "environment": {
    "locked": true,
    "operator_notes": "FILL IN (date; anything unusual during the run)"
  }
}

```

```

},
"spectrometer": {
  "probe_type": "RT",
  "console": "Agilent DD2 400",
  "probe_string": "5 mm HCN",
  "coil_temp_k": 296.5,
  "preamp_temp_k": 296.5
},
"instrument": {
  "vnmrj_version": "3.2 Revision A",
  "spectrometer_model": "Agilent DD2 400",
  "field_state_notes": "FILL IN (lock on/off during the noise blocks; z0 stable?)"
},
"calibration": {
  "p90_us": 10.0,
  "p90_power_db_or_w": "FILL IN (tpwr, dB)",
  "topshim_ok": false
},
"experiments": [
  { "expno": 2000, "tuning": { "setting_index": 0, "label": "FILL IN (normal tuning)",
    "tune_reading": null, "match_reading": null, "units": "FILL IN" } },
  { "expno": 2001, "tuning": { "setting_index": 0, "label": "FILL IN (normal tuning)",
    "tune_reading": null, "match_reading": null, "units": "FILL IN" } },
  { "expno": 2010, "tuning": { "setting_index": 1, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2011, "tuning": { "setting_index": 1, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2020, "tuning": { "setting_index": 2, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2021, "tuning": { "setting_index": 2, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2030, "tuning": { "setting_index": 3, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2031, "tuning": { "setting_index": 3, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2040, "tuning": { "setting_index": 4, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } },
  { "expno": 2041, "tuning": { "setting_index": 4, "label": "FILL IN", "tune_reading": null,
    "match_reading": null, "units": "FILL IN" } }
]
}

```

Three mechanical notes. The two temperatures are the physical lab ambient you read in 2.1 — replace 296.5 with your reading, or `null` if you have none. Nothing else needs listing: the converter reads each experiment’s role from its save name (`_sn_ladder`, `_sn_ref_open`, `_sn_noise`, `_sn_ref_close`, `_sn_tune`, `_sn_signcal`) and builds the gain ladder from the 27 rungs’ stored gains, so the questionnaire no longer lists noise blocks or ladder rungs. The `tuning` entries are the one thing only you know: setting 0 is your normal tuning, `tune_reading` and `match_reading` are the numbers the console showed at that setting (or `null`), `units` names them (“turns”, “% reflected”, “dB”), and `label` is free text; add a line per block for settings beyond 4 or for blocks beyond the second (20k2 to 20k5 on a 1%-class tube).

3.3 Pack, check, upload

Three commands, run from inside the unzipped `spin-noise-network` folder:

```
python3 packer/pack_bundle.py --vendor agilent --answers answers.json
```

```
/path/to/the/data/folder
```

Success looks like: the packer validates the session and prints the path of one `spinnoise_siu-carbondale_*.zip` — and if anything is missing, it lists exactly what, in plain text, so you fix it and re-run.

```
python3 uploader/upload_bundle.py --selftest spinnoise_*.zip
```

Success looks like: the last line reads `RESULT: PASS` — entirely local, no network involved yet.

```
python3 uploader/upload_bundle.py spinnoise_*.zip
```

Success looks like: the last line reads `RECEIPT: <id>` — the server has your bundle, checksum-verified on arrival; uploads are chunked and resumable, so if the connection drops, re-run the same command and it picks up where it left off.

4 If anything goes wrong

One rule: send it, don't fix it

Send the **exact error text** (copy-paste, not a paraphrase) and the file it choked on — the offending `NN_*.fid` folder zipped, or the bundle zip — to `jwbquantum@gmail.com`. The converter has met one real Agilent instrument — this one, two sessions — and every further session is how it becomes trustworthy beyond that. Nothing you can do here damages anything — the software only reads your data, the uploader never deletes your zip, and the worst case at every step is re-running one command. If the whole chain fights you, zip the entire session directory and e-mail that (a share link is fine if it is large); we take it from there.

Contact: John W. Blanchard (`jwbquantum@gmail.com`) — CASPER@night software v0.7.1 — September 16, 2026. Your raw data remain yours, and contributing facilities are co-authors on network publications their data enter.